

Salah Boubnider University

Academic Year : 2025 - 2026

Faculty of Medicine of Constantine

Cytology Courses

## **The Interphase Nucleus and Cell Cycle Control**

**I. General Concepts:** The nucleus is a large cellular organelle, generally ovoid in shape. It was first identified by the Scottish botanist Robert Brown in 1831, in plant cells. The nucleus contains more than 99% of the cellular DNA (the remainder is contained within the mitochondria).

### **I.1.Nuclear size:**

It varies according to the cell type and its activity (5 to 6  $\mu\text{m}$  in diameter); it represents approximately 3 to 10% of the cell diameter (20 to 25% of the cellular volume).

### **I.2. Number of nuclei:**

Generally, there is a single nucleus per cell, but there are:

- ✓ **Anucleated cells (without a nucleus):** Red blood cells (Erythrocytes) and Corneocytes.
- ✓ **Binucleated cells:** Hepatocytes, cardiac cells, urinary epithelium, cartilage.
- ✓ **Multinucleated cells:** Myocytes (muscle cells), osteoclasts (bone cells).

### **I.3. Nuclear shape:**

It differs depending on the cell's morphology and activity (**Fig. 1**):

- ✓ **Spherical shape:** Lymphoid cells, neurons, hepatocytes...
- ✓ **Elongated shape:** Muscle cells, fibroblasts...
- ✓ **Polylobed shape:** Polymorphonuclear cells.
- ✓ **Ovoid shape:** Columnar cells of the intestinal epithelium.

## **II. Structure and ultrastructure of the nucleus:**

The nucleus exhibits a highly complex organization and structure; it contains the following elements:

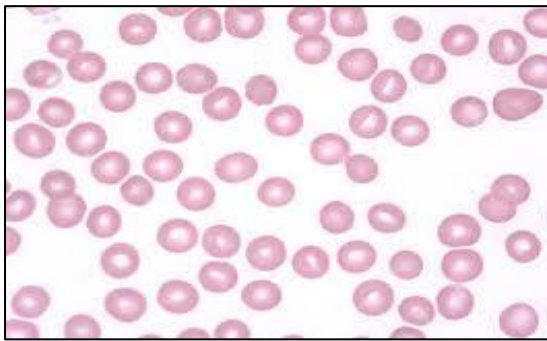
### **II.1. The nuclear envelope:**

It delimits the nucleus by separating it from the cytoplasm. It is composed of two phospholipid membranes, inner and outer, which control the exchanges between the nucleoplasm (intranuclear medium) and the cytoplasm through nuclear pores (**Fig. 2**).

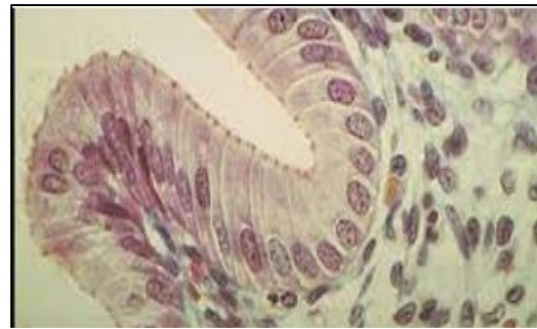
**A. The outer nuclear membrane:** Continuous with the ER (endoplasmic reticulum) membrane, it is studded with ribosomes and participates in protein synthesis. It carries transmembrane proteins (Nesprins) that anchor the nuclear envelope to the cytoplasmic

cytoskeleton (**Fig. 3**). transmembrane proteins (**Nesprins**) that anchor the nuclear envelope to the cytoplasmic cytoskeleton (**Fig. 3**).

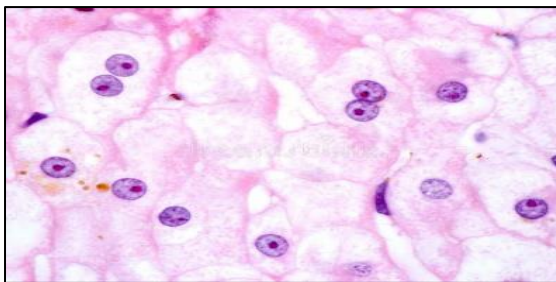
- **B. The inner nuclear membrane:** Lined by the **lamina** (a fibrous protein network), it provides structural support. Its transmembrane proteins (**SUN**) bind lamins and chromatin (via histones), forming a bridge with Nesprins to link the lamina to the cytoskeleton. Both membranes define the perinuclear space, which is in direct continuity with the lumen of the ER (**Fig. 3**).



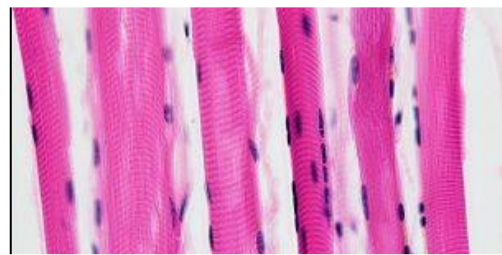
Red blood cells (enucleated cells)



Cylindrical cells (mononucleated, ovoid nucleus).

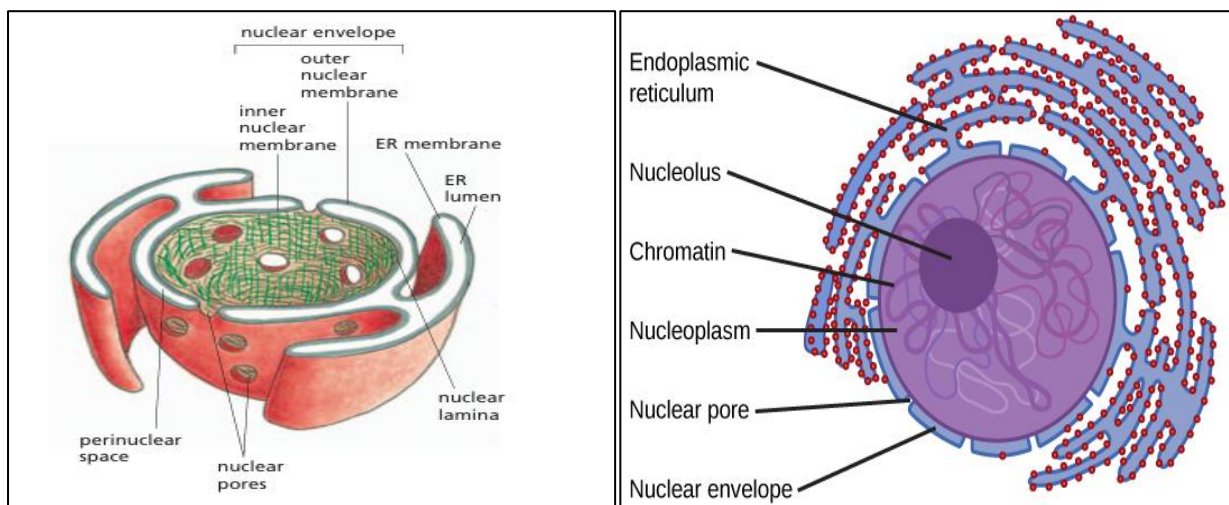


Hepatocytes (binucleated, spherical nucleus)

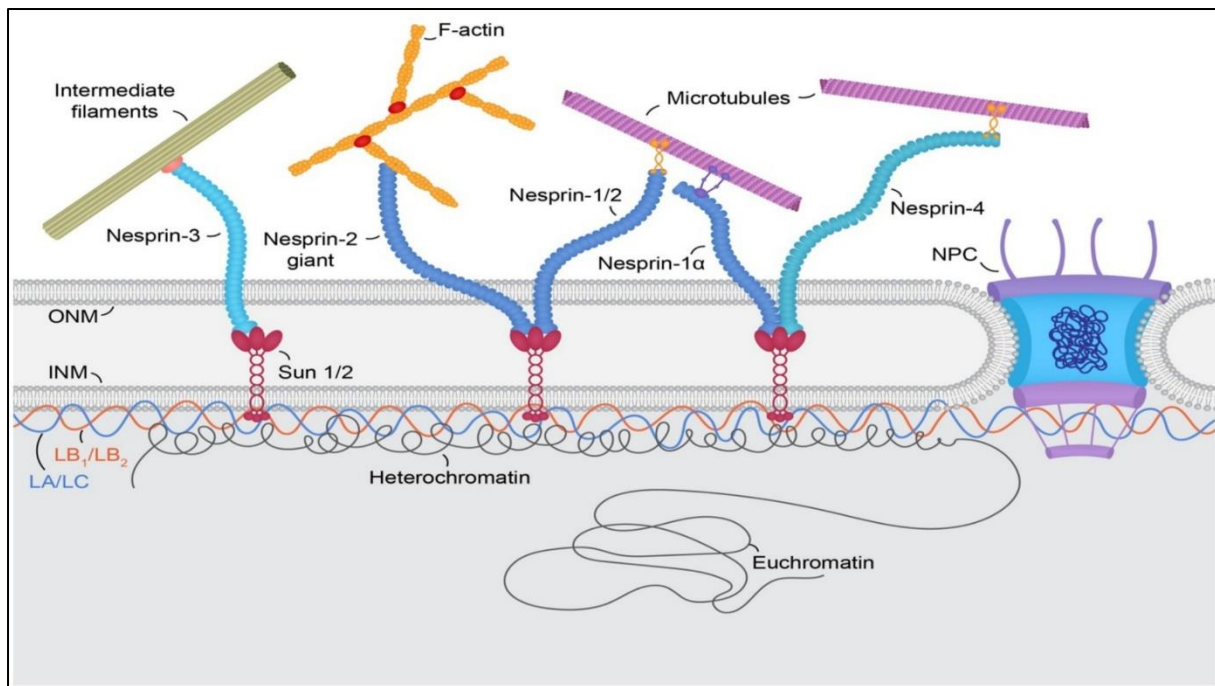


Myocytes (multinucleated, elongated shape)

**Figure 1.** Various types of cell nuclei.



**Figure 2.** Schematic representation of the nucleus.



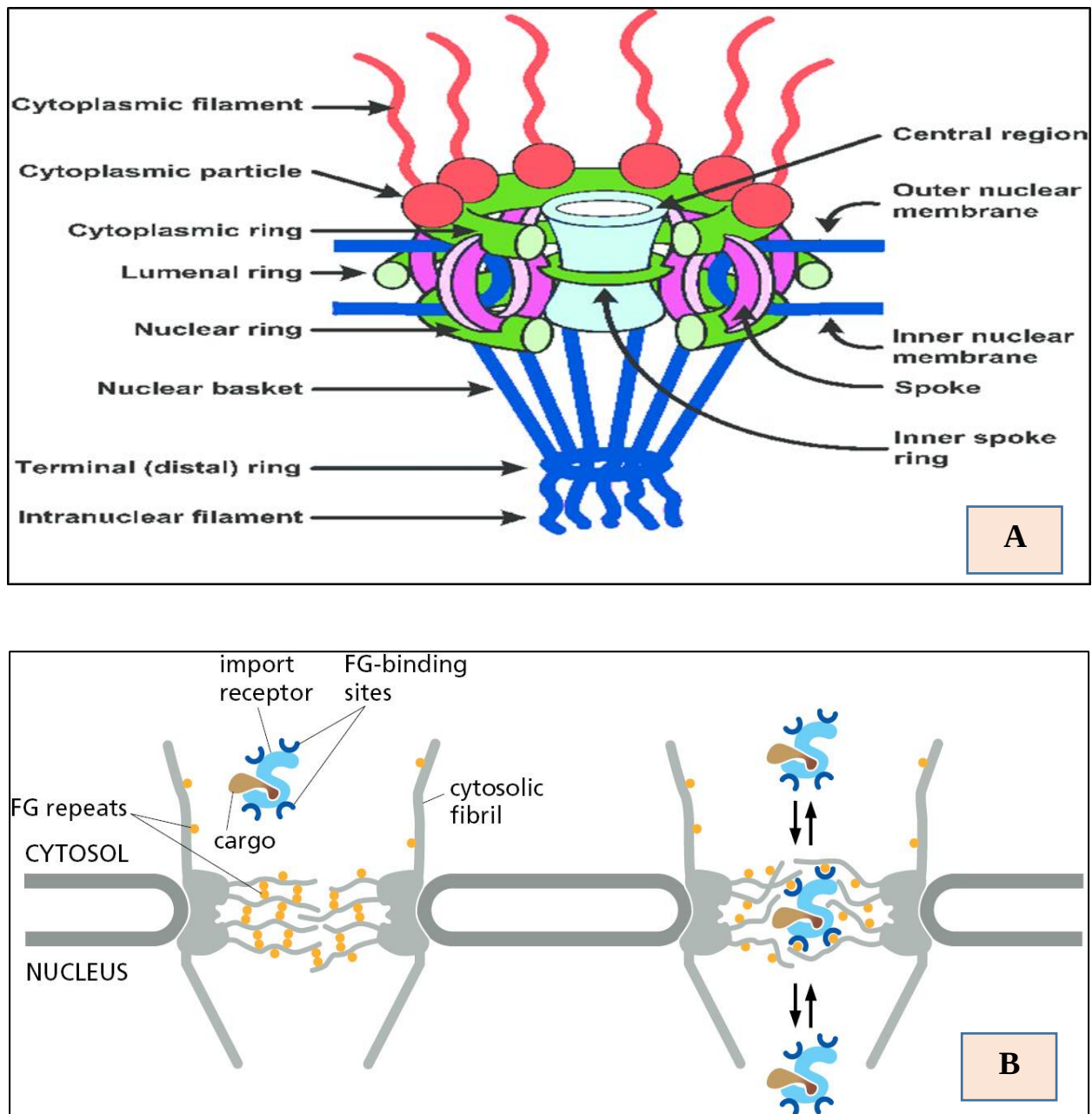
**Figure 3.** Molecular organization of the nuclear envelope. ONM: Outer nuclear membrane. INM: Inner nuclear membrane. NPC: Nuclear pore complex.

**II.2. Nuclear pores:** These are protein complexes organized into channels (**Fig. 4**). Their role is to ensure molecular exchange between the cytoplasm and the nucleoplasm (RNA, proteins, ribosomal subunits, nucleotides, etc.); their total diameter is approximately 120 nm (**Fig. 5**).

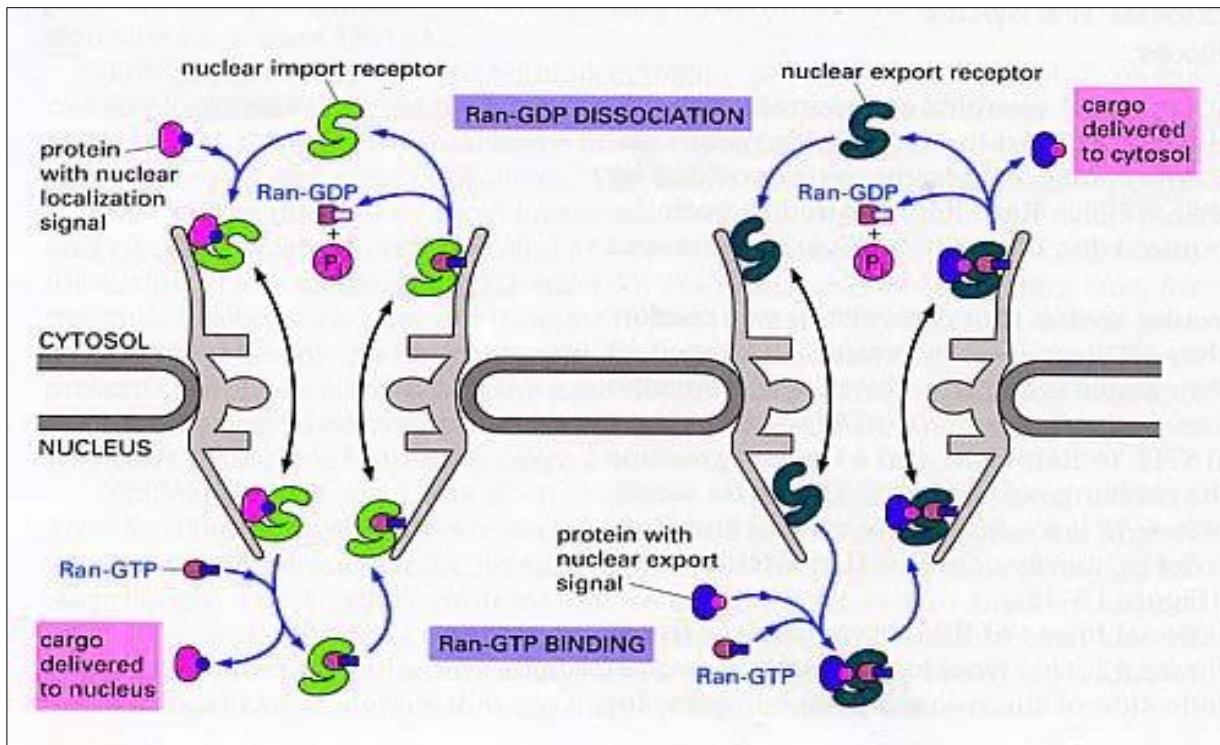
**II.3. The nucleoplasm (intranuclear medium):** This is a gelatinous matrix containing chromatin, ions, proteins, enzymes, nucleotides, and other components.

**II.4. The nucleolus:** This is a nuclear compartment not delimited by a membrane (**Fig. 6**), responsible for the synthesis of ribosomal RNA (rRNA) and the assembly of ribosomal subunits. The biochemical components of the nucleolus include:

- ✓ **Ribosomal DNA:** 200 copies of the genes encoding three rRNAs of nucleolar origin. These copies are distributed across the short arms of five pairs of chromosomes (chromosomes 13, 14, 15, 21, and 22). Each chromosome contains 20 copies of the genes encoding these rRNAs.
- ✓ **Ribosomal RNAs:** The rRNAs synthesized within the nucleolus are the 28S, 5.8S, and 18S rRNAs, whereas the 5S rRNA is synthesized outside the nucleolus, though still within the nucleus.
- ✓ **Proteins:** These represent approximately 85% of the nucleolus's mass. They primarily consist of enzymes, such as RNA polymerase I, as well as numerous ribosomal proteins that assemble with rRNAs to form ribosomal subunits.



**Figure 4.** Molecular organization of a nuclear pore (A). Mechanism of selective transport through the nuclear pore complex (B). **FG repeats:** These are repeated sequences of Phenylalanine (F) and Glycine (G). These motifs are present throughout the channel and form a selective filter for large molecules ( $\geq 40$  kDa). They are found on the cytosolic filaments (to capture molecules at the entrance), inside the central channel (where they form a dense network or "gel"), and on the nuclear filaments (to guide the exit). **Cargo:** Is the molecule to be transported.



**Figure 5.** Transport of large molecules through the central transporter of the nuclear pore.

**Left:** During the import of molecules from the cytosol to the nucleus, importin binds to the NLS (Nuclear Localization Signal) of the cargo molecule. The importin-cargo complex crosses the nuclear pore. In the nucleoplasm, Ran-GTP interacts with the importin, triggering the release of the imported molecule. The return of the free importin to the cytosol requires Ran-GTP.

**Right:** During the export of molecules to the cytosol, exportin recognizes the NES (Nuclear Export Signal) of the cargo molecules. Export is facilitated by the binding of Ran-GTP to the exportin-cargo complex. In the cytosol, the exported molecules are released following GTP hydrolysis.

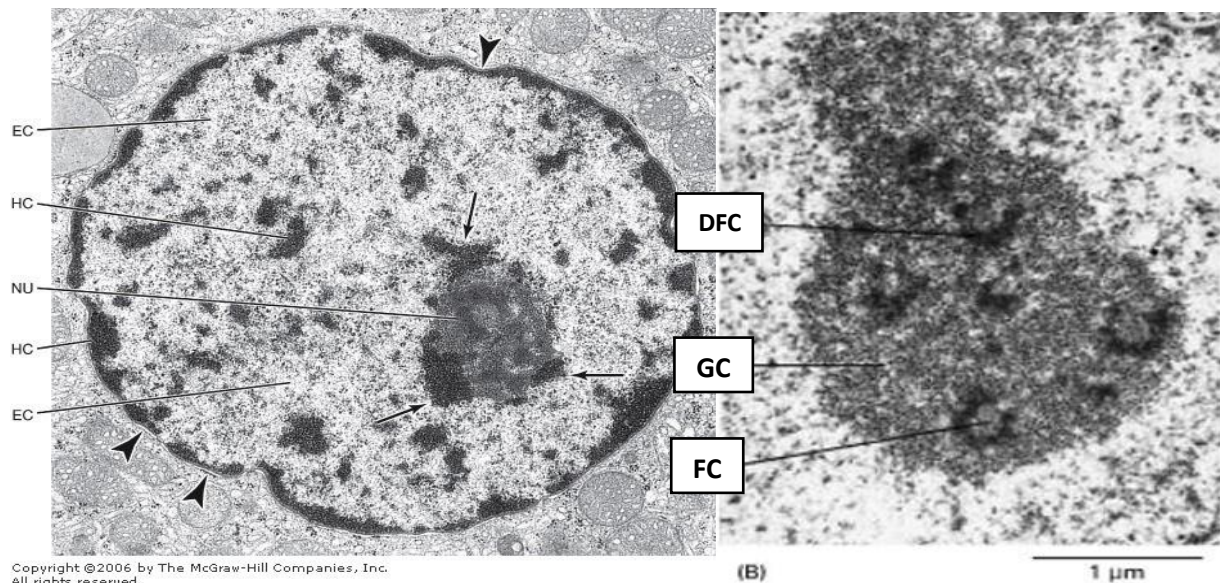
**Important:** mRNA does not have an "NES" sequence. It is recognized by its complete maturation (5' Cap + Splicing + Poly-A tail), then exported by NXF1/NXT1 without the direct involvement of the Ran-GTP cycle.

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**II.5. Chromatin:** This is the nuclear genetic material consisting of DNA strands arranged in a double helix and DNA-bound proteins (histones). Chromatin can be divided into two types:

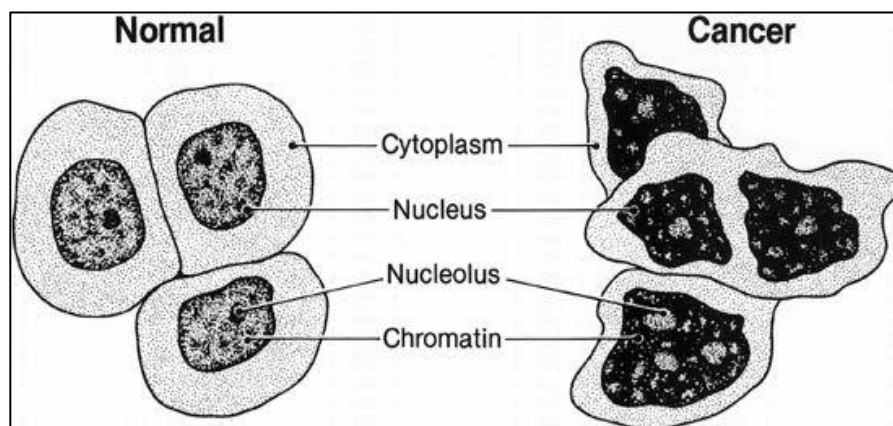
**A. Euchromatin:** This is the decondensed form of chromatin during interphase (very lightly stained). It corresponds to active DNA regions (high transcriptional activity).

**B. Heterochromatin:** This is the condensed form of chromatin during interphase. It corresponds to inactive DNA regions (transcriptionally silent). (**Fig. 6**)



**Figure 6.** Ultrastructure of the nucleus (A) and nucleolus (B) observed by transmission electron microscopy (TEM). EC: Euchromatin; HC: Heterochromatin; NU: Nucleolus; **FC**: Fibrillar center: Zone containing ribosomal DNA (rDNA); **DFC**: Dense fibrillar component: Zone surrounding the FC, containing rRNAs being transcribed (primary transcripts); **GC**: Granular component: Peripheral zone containing ribosomal subunits being assembled (RNA + proteins). This is the site of final maturation.

- ❖ **Nuclear Abnormalities and Cytopathological Diagnosis:** The nucleus is the subject of detailed morphological analysis, particularly in cancer diagnosis. The following figure summarizes the morphological differences between a normal cells and a cancer cells:



**Figure 7.** Normal cells possess a large cytoplasm and a single nucleus with a single nucleolus and fine chromatin. In contrast, cancer cells exhibit a small cytoplasm, enlarged (and sometimes multiple) nuclei with large, multiple nucleoli, as well as coarse chromatin (heterogeneous and clumped appearance).

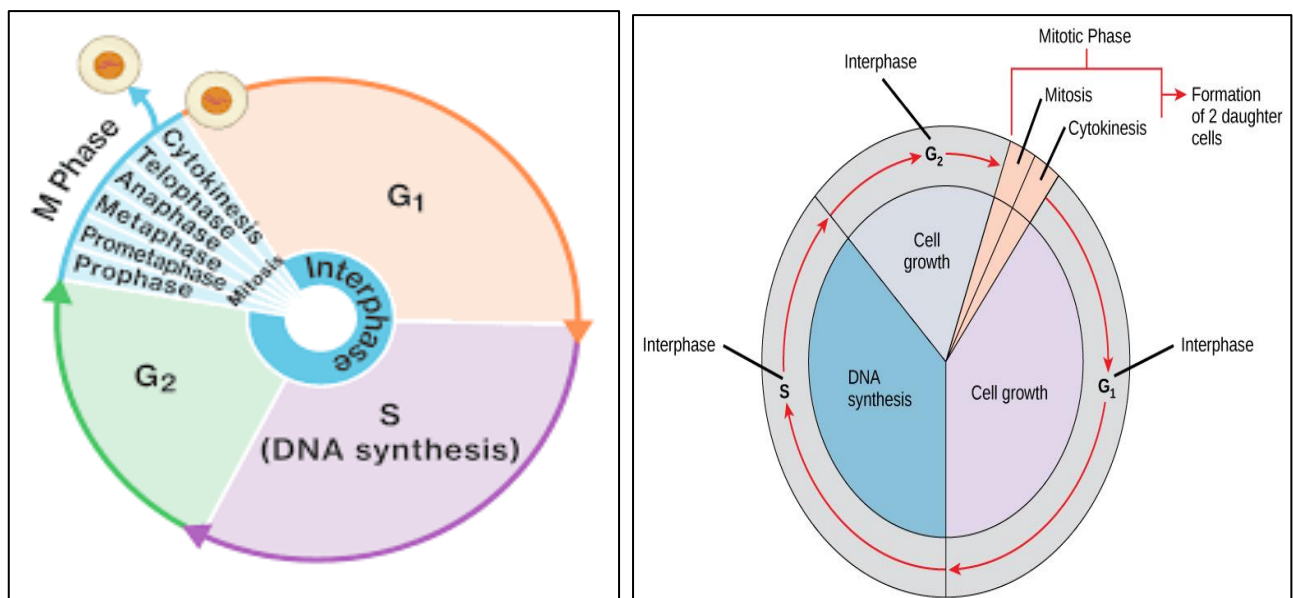
### III. The Cell Cycle

**Definition:** The cell cycle refers to the sequence over time of two main phases in cellular life: **interphase**, during which the cell grows and duplicates its genetic material, and the **cell division phase (M phase)**, during which the cell divides to produce two identical daughter cells.

**III.1. Interphase:** This is the period between the end of one division (mitosis) and the beginning of the next, typically representing the largest portion of the cell cycle. It comprises the **G<sub>1</sub>, S, and G<sub>2</sub> phases**.

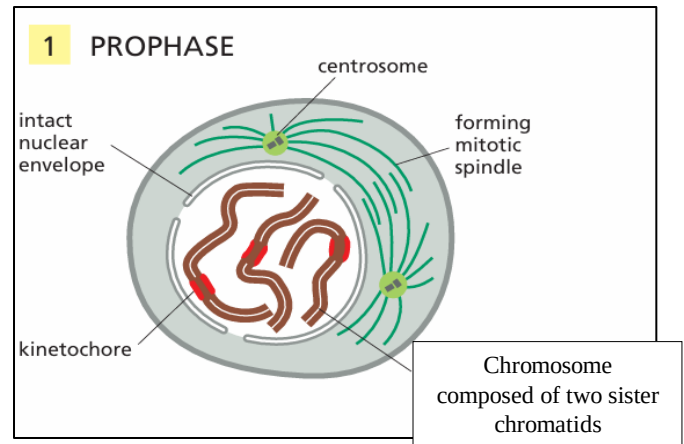
- **a. G<sub>1</sub> phase (G: Gap):** This is the longest phase of the cell cycle and the most variable depending on the cell type. This phase is characterized by growth and the synthesis of proteins necessary for the S phase.
- **b. S phase (S: Synthesis):** This is a short phase during which **DNA replication** occurs. The amount of nuclear DNA is doubled by the end of this phase.
- **c. G<sub>2</sub> phase:** This is the second growth phase. It begins once DNA replication is complete. During this phase, the cell contains twice the usual amount of DNA and prepares for mitosis by synthesizing all necessary components.

**III.2. M Phase (Mitotic Phase):** It includes **mitosis** (divided into 5 stages: prophase, prometaphase, metaphase, anaphase, and telophase) and concludes with **cytodieresis or cytokinesis** (the division of the cytoplasm).

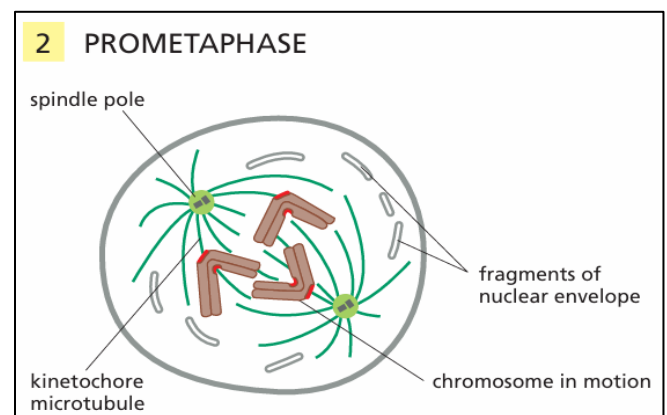


**Figure 8.** Cell cycle phases.

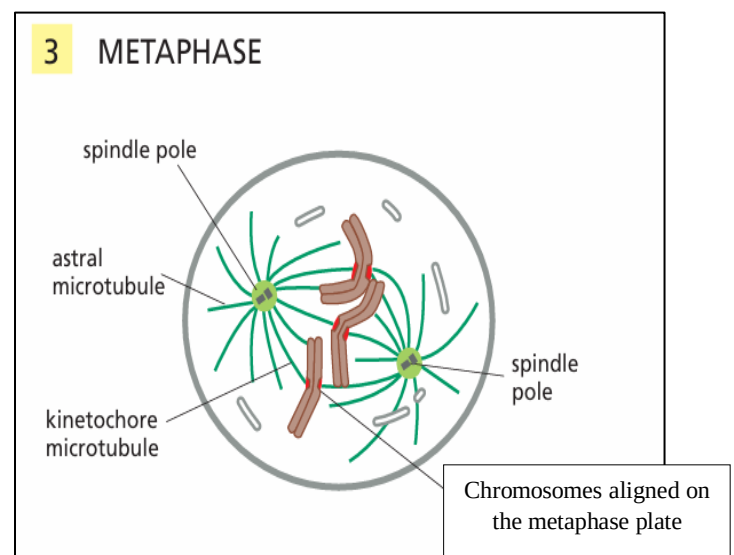
**1. Prophase:** During prophase, the duplicated chromosomes condense; each consists of two sister chromatids tightly associated at the centromere. Outside the nucleus, the mitotic spindle assembles between the two centrosomes, which have begun to move apart. The Golgi complex and the ER fragment into small vesicles. For simplicity, only three chromosomes are



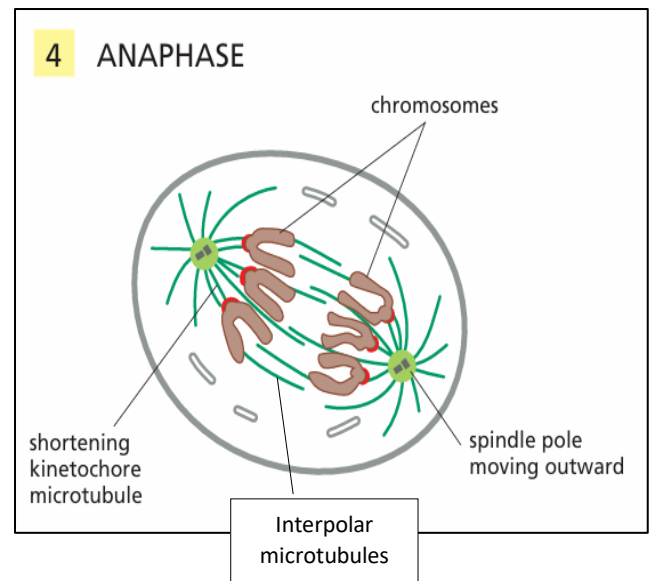
**2. Prometaphase:** It begins with the fragmentation of the nuclear envelope into small vesicles (following the phosphorylation of lamins and nuclear pore proteins). The chromosomes can then attach, via their kinetochores, to the kinetochore microtubules of the mitotic spindle, which guide them toward the cell equator.



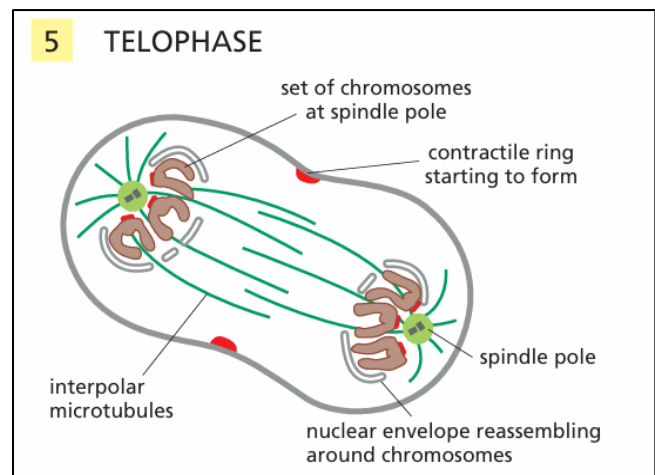
**3. Metaphase:** During this phase, the chromosomes align precisely along the metaphase plate (equatorial plate). The kinetochore microtubules link each sister chromatid to an opposite pole, while the astral microtubules stabilize the entire structure. This rigorous organization ensures that the genetic material will be equally distributed between the two future daughter cells.



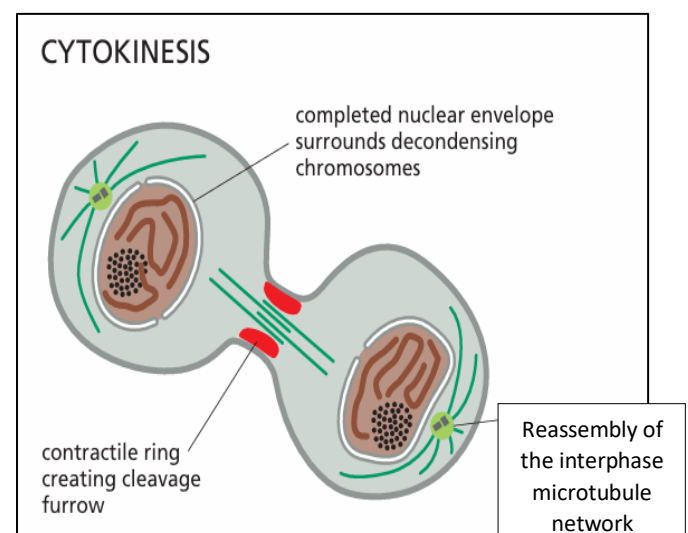
**4. Anaphase:** It begins with the abrupt degradation of centromeric cohesin (the protein complex binding the two chromatids) by separase, allowing the separation and migration of sister chromatids toward opposite poles. The separated chromatids are slowly pulled toward the cell poles by the shortening of kinetochore microtubules. Meanwhile, astral microtubules help stabilize and orient the structure outward. Simultaneously, the interpolar microtubules slide past one another to push the poles apart and elongate the cell.



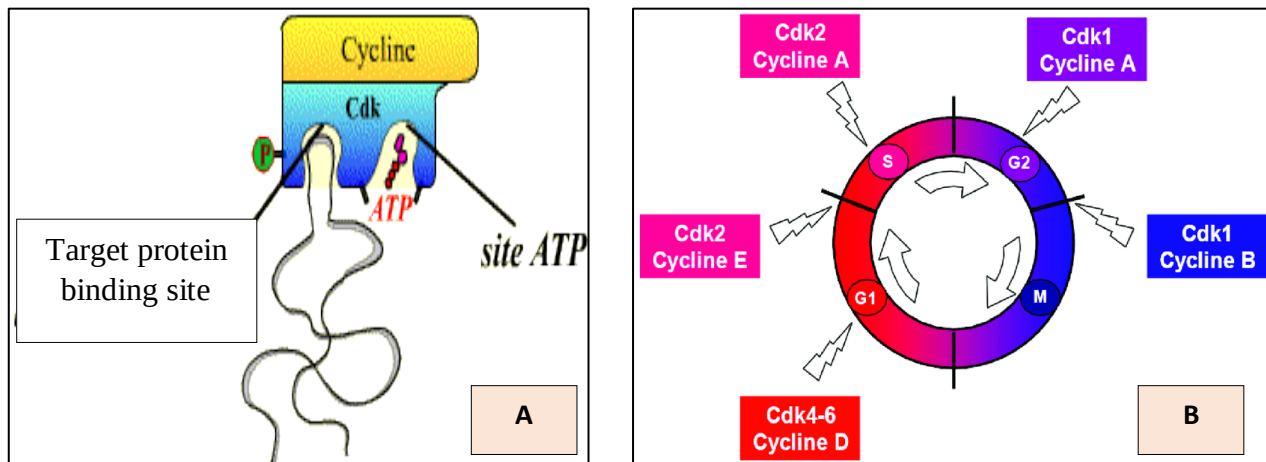
**5. Telophase:** The two sets of chromosomes are now gathered at the spindle poles, where they begin to decondense. Around each set, a new nuclear envelope reforms, thus isolating the genetic material within two distinct future nuclei. Meanwhile, the interpolar microtubules remains clearly visible: they continue to slide past one another to maintain pole separation and further elongate the cell. Finally, a contractile ring begins to assemble at the center of the cell.



**Cytokinesis:** This is the final stage of cell division, where the cytoplasm is divided between the two new daughter cells. This phenomenon is driven by the contractile ring which, by tightening, creates a cleavage furrow on the cell surface. Simultaneously, normal cellular life resumes: the nuclei are now fully reformed, the chromosomes decondense back into chromatin, and a new interphase begins.



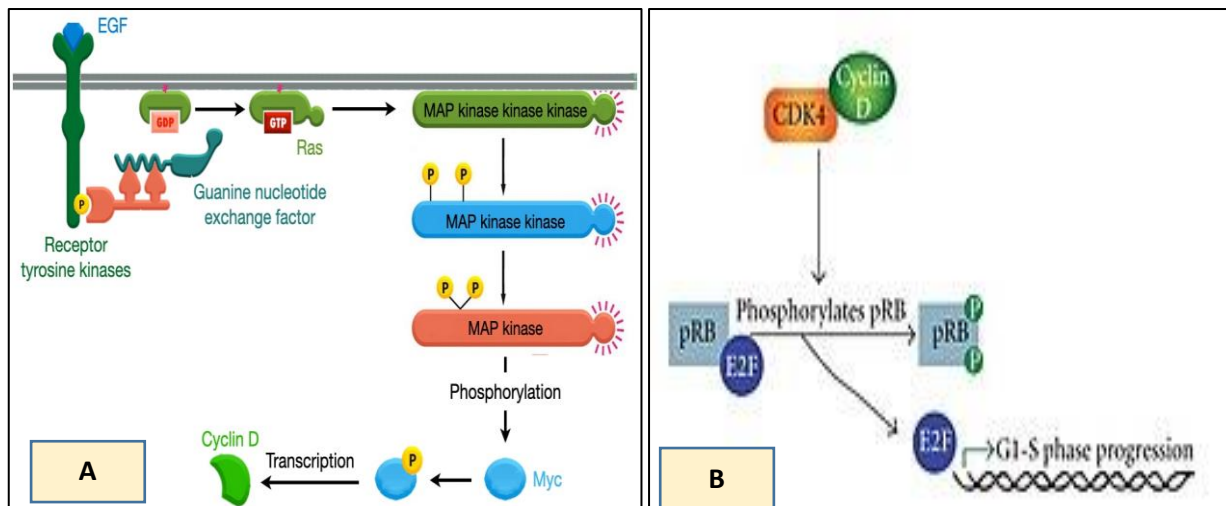
**II.1. Molecular mechanisms of cell cycle control:** Cell cycle progression is regulated by the formation and sequential activation of specific Cyclin-CDK complexes, whose activity coordinates the transition from one phase to the next (**Fig. 9**).



**Figure 9.** Molecular mechanism of Cyclin-CDK complex function (A). Sequential regulation of the cell cycle by Cyclin-CDK complexes (B).

**Tableau 01 :** Tableau récapitulatif des moteurs moléculaires du cycle cellulaire.

| Phase         | Main Complex          | Biological Role                                                        | Key Molecular Actions                                                |
|---------------|-----------------------|------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>G1</b>     | Cyclin D / CDK4-6     | <b>Division decision:</b> Environmental analysis (nutrients, signals). | Phosphorylates <b>Rb protein</b> to release <b>E2F</b> .             |
| <b>G1 / S</b> | Cyclin E / CDK2       | <b>Irreversible commitment:</b> Crossing the "point of no return".     | Activates <b>helicases</b> to unwind the DNA double helix.           |
| <b>S</b>      | Cyclin A / CDK2       | <b>Genomic integrity:</b> Duplication of the DNA stock.                | Activates <b>DNA polymerases</b> and the proofreading mechanism.     |
| <b>S / G2</b> | Cyclin A / CDK1       | <b>G2 Transition:</b> End of replication and start of condensation.    | Initiates <b>chromosome condensation</b> before mitosis.             |
| <b>G2 / M</b> | Cyclin B / CDK1 (MPF) | <b>Equitable distribution:</b> Organization and execution of division. | <b>Nuclear envelope breakdown</b> , spindle assembly, and alignment. |



**Figure 10.** From extracellular signaling to S-phase commitment.

**(A)** Cycle entry begins with the recognition of a growth factor by its membrane receptor. This signal triggers a cascade of intracellular reactions resulting in the synthesis of Cyclin D, a key protein that initiates G1 phase progression.

**(B)** To become active, the Cyclin D / CDK4 complex must be phosphorylated by the CAK complex (Cyclin H / CDK7). Once operational, it phosphorylates the **Rb** protein, thereby releasing the E2F transcription factor. E2F then activates the expression of S-phase genes, marking the cell's definitive commitment to division.

❖ **Mitotic Spindle Inhibitors:** Successful division depends on microtubule dynamics—the ability to both assemble and disassemble. Two families of therapeutic molecules disrupt this mechanism:

- ✓ **Vinca Alkaloids** (e.g., Vinblastine): They prevent spindle formation (inhibition of polymerization). Chromosomes remain scattered, blocking the cell in prophase/prometaphase.
- ✓ **Taxanes** (e.g., Taxol): They prevent spindle breakdown (inhibition of depolymerization). The spindle becomes "frozen," which blocks chromosome migration toward the poles and maintains the cell in metaphase.
  - ✓ **Consequence:** These agents activate the M-phase checkpoint (Spindle Assembly Checkpoint). The cell, unable to proceed to anaphase, ultimately dies by apoptosis.